



Microbiota changes in rumen and milk corresponding to dietary protein intake in transition dairy cows

J. Scott, , J. S. Brouard, G. Drouin, , D. R. Ouellet, , C. Ster, , and R. M. Petri* 

Sherbrooke Research and Development Centre, Agriculture and Agri-Food Canada, Sherbrooke, QC, J1M 0C8, Canada

ABSTRACT

During the transition period in dairy cows, the incidence of disease increases due to a negative energy balance affecting both the metabolic and immune health status. Limiting milk production at the beginning of lactation improves the metabolic status of cows. However, past strategies tested to achieve this reduction either negatively affected milk yield for the rest of the lactation or were difficult to implement on large-scale dairy farms. This study evaluated the impact of a temporary reduction in MP supply during the transition period on the rumen and milk microbiota and their metabolic composition. Treatment cows ($n = 5$) were fed 80% of their MP needs (MP80) from 14 d before calving to 14 d after calving, before being switched to a 100% MP diet (MP100) for an additional 14 d. Control cows ($n = 6$) were fed MP100 for the entire experiment. Samples of rumen content and milk were taken in the immediate postpartum phase (PP) on d 2 and 7, as well as after dietary change in the experimental recovery phase on d 21 and 28 postpartum. All samples were extracted for DNA and analyzed using shotgun metagenomic sequencing (Illumina NovaSeq). Milk samples were additionally analyzed for composition, and rumen fluid was analyzed for short-chain fatty acids and ammonia-N. Significant changes to the microbial composition were almost exclusively associated with the effect of day of sampling, with the exception being the family *Micrococcaceae*, which was found to be differentially abundant in the MP100 compared with the 80% group in PP milk samples. This study used a metagenomics approach to understanding the impact of altered protein supply on rumen and milk microbiota, to better understand impacts on these separate ecosystems.

Key words: rumen and milk microbiota, transition dairy cows, metabolizable protein, shotgun metagenomics

INTRODUCTION

During early lactation, dairy cows experience a state of negative energy balance (NEB) due to the lag in feed intake relative to the escalating energy demands of milk synthesis (Bauman and Currie, 1980). As milk production is a metabolic priority, cows mobilize body reserves, primarily in the form of adipose tissue and muscle protein, to compensate for this energy deficit (McArt et al., 2013). Therefore, it is critical to optimize dietary strategies to alleviate NEB and support the effective transition of dairy cows for their overall health and milk production efficiency.

A key factor in meeting the cow's energy demands is the role of rumen microbes, which facilitate the fermentation of dietary substrates to produce short-chain fatty acids (SCFA), the primary energy source for the host (Bergman, 1990). Dietary changes influence the rumen microbiota, which in turn affects the energy and protein supply for the host animal. Therefore, transition periods, with their dietary adaption, can influence the composition and function of the rumen microbiota. The efficiency of microbial fermentation directly influences the cow's overall energy balance and, consequently, its milk production potential (Van Soest, 1994). As such, energy-rich diets enhance microbial protein yield and are directly correlated with increased milk yield and milk protein (Zhou et al., 2015). However, energy-rich diets can also negatively affect the rumen microbial composition and function due to rapid and sustained fermentation combined with a lag in the absorptive capacity of the rumen, leading to lower ruminal pH, ruminal dysbiosis, decreased feed intake, and a subsequent decrease in milk production (Plaizier et al., 2018).

In addition to energy, protein intake is a crucial dietary component influencing metabolic maintenance, milk protein synthesis, and milk quality. Rumen microbes

Received September 9, 2025.

Accepted February 28, 2026.

*Corresponding author: renee.petri@agr.gc.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

synthesize microbial protein from dietary protein, which is a key source of amino acids for milk production (Palmonari et al., 2024). Feeding high-protein diets is commonly done to maximize efficiency and milk protein yields in lactating dairy cows. However, supplementing protein over the metabolic capacity of the animal during transition results in decreased N utilization efficiency, increased N excretion in manure, and a reduced overall profitability for dairy farmers (Zang et al., 2021). Unlike energy-rich diets, protein-rich diets do not markedly affect microbial protein yield or affect the abundance of microbial genera involved in energy metabolism (Belanche et al., 2012; Jami et al., 2014). However, efficient microbial protein synthesis depends on optimal rumen function and dietary balance (Brassard et al., 2015). Despite important advancements in transition nutrition, current dietary strategies remain insufficient to fully mitigate NEB during the transition period, therefore alternative methods are required to support transition cows.

Recently, studies have shown that reducing protein intake may optimize microbial efficiency and energy utilization, thereby reducing metabolic stress, improving nitrogen efficiency, and supporting transition cow health (Rehman et al., 2020; Leduc et al., 2023). In addition to this, other studies have highlighted possible correlations between the rumen microbiota and the milk microbiota under various dietary challenges (Jami et al., 2013; Lima et al., 2015). Therefore, the composition and functionality of these microbial communities could be simultaneously influenced by dietary changes (Pitta et al., 2014), which may, in turn, affect milk production efficiency and milk composition (McGarvey et al., 2004). Understanding these connections is essential for developing new strategies to enhance the ability of a dairy cow to navigate the transition period effectively using a whole-systems approach to nutrition.

This study aims to explore the relationships between dairy cow energy requirements, the role of rumen microbes in energy provision, and the effects of protein supply on milk quality, with a focus on the interactions between rumen and milk microbiota under temporarily reduced MP supply. We hypothesize that reducing MP during early lactation enhances rumen microbial efficiency and energy utilization, resulting in improved nitrogen efficiency in transition cows. Additionally, we propose that restoring full MP recommendations at 2 wk postpartum, in combination with increased DMI, will facilitate the transition of rumen microbiota and milk microbiota to their full metabolic capacity, leading to an increase in milk production and improved postpartum health. By integrating rumen and milk microbiota, this study aims to refine nutritional strategies to enhance dairy cow health and optimize milk production.

MATERIALS AND METHODS

Experimental Design

Using the ARRIVE reporting guidelines (Percie du Sert et al., 2020), the authors conducted this experiment in accordance with the guidelines of the Canadian Council on Animal Care (1984) and it was approved by the Sherbrooke Research and Development Centre Animal Care Committee (no. 591). As visualized in Figure 1, 11 cannulated pregnant multiparous Holstein cows were enrolled from d -28 (before anticipated calving; d 0) to 28 d postpartum. Selection criteria and reasoning for sampling numbers were previously described (Tapp et al., 2025). Animals were maintained in a tiestall facility, except around calving (calving pen), were fed a TMR twice daily at 0915 and 2100 h, and were given ad libitum access to water. Animals were randomized into one of 2 groups. Cows receiving reduced protein ($n = 5$) were provided a ration formulated to supply 80% of their MP (**MP80**) requirements starting at d -14 before calving until d 14 after calving; then they were transitioned to a ration formulated to supply 100% of their MP (**MP100**) requirements from d 15 to 28 as a recovery period. The control cows ($n = 6$) received diets formulated to fulfill their MP requirements during the precalving period and through lactation, based on a projected milk yield of 40 kg/d. Diets were formulated using the CNCPS model (v. 6.55; Van Amburgh et al., 2015) and have been previously reported (Tapp et al., 2025). All rations were isoen-ergetic, and feed intake was monitored daily. Sampling was divided into 2 phases, the postpartum phase (**PP**) accounting for sampling done on d 2 and 14, and the recovery phase (**RP**) on d 21 to 28.

Sample Collection, Preservation, and Preparation

During the entire experiment, DMI was recorded daily. Rumen samples were taken on d 2, 7, 21, and 28 after calving at 2 h after the morning meal using sterilized collection containers and clean gloves for each individual animal. Samples of approximately 150 g of whole rumen content were collected from each of 5 points within the rumen (cranial-ventral, cranial-dorsal, caudal-ventral, caudal-dorsal, and medial) for each animal. After homogenization, a subsample was taken for microbial analysis and immediately stored at -80°C . The remainder of the rumen content was strained through cheese cloth and the liquid was immediately analyzed for pH (SevenExcellence model S400 pH meter, Mettler Toledo, CA), whereas a subsample for NH_3 was stabilized with 2 mL of 6 *N* hydrochloric acid and then stored at -20°C . The NH_3 analysis was based on the method described by Weatherburn (1967) with colorimetric analysis con-

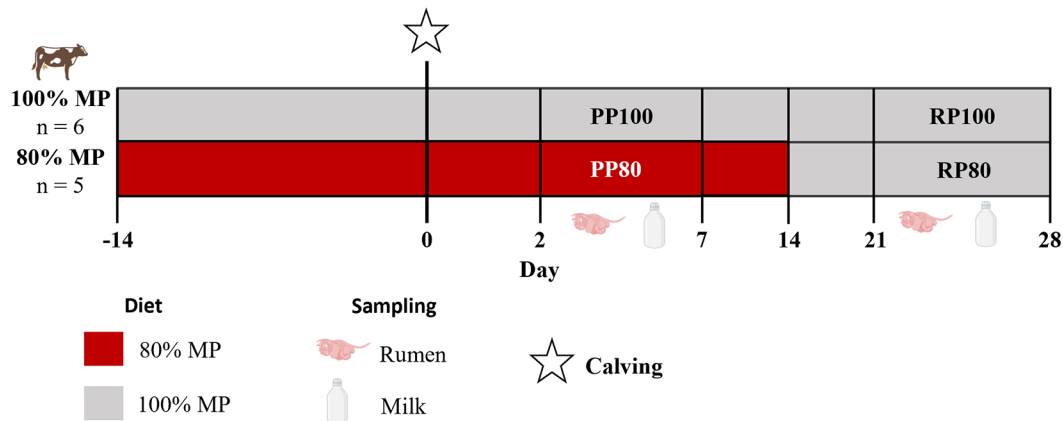


Figure 1. Schematic representation of the experimental design for the animal trial showing from 14 d before calving (d -14), to 28 d after calving (d 0), and then days of sampling. Red bars indicate period of MP restriction; gray bars indicate feeding of full MP at recommended levels. PP100: animals fed 100% of recommended MP during the first 2 wk postparturition; PP80: animals fed 80% of recommended MP during the PP; RP100: animals fed MP100 during the RP; RP80: animals with dietary change from 80% MP, then converted to MP100 on d 14 and sampled on d 21 and 28 in the RP.

ducted on the centrifuged supernatant at 625 nm using a spectrophotometer (SpectraMax ABS, Molecular Devices, Downingtown, PA). Final NH_3 concentrations (mmol/mL) were calculated using a standard dilution curve with ammonium sulfate.

All cows were milked twice daily at 0700 and 1900 h. Milk production was monitored at each milking and aseptic samples were collected at morning milking on d 2, 7, 21, and 28. Subsamples were preserved in bronopol and stored at 4°C before being sent to Lactanet (Canadian Network for Dairy Excellence, Ste-Anne-de-Bellevue, Canada) for determination of lactose, protein, and fat content, and SCC as well as the fatty acid (FA) profile. Additional subsamples were stored at -80°C for DNA extraction.

DNA Extraction of Rumen and Milk

The whole rumen content samples were thawed and then mixed to ensure uniformity; then, 10 to 15 g of the mixture was aliquoted into Whirl-Pak bags (Nasco, WI) and methylcellulose release buffer was added at a 2:1 ratio to sample. The sample mixture was homogenized twice at 260 rpm for 90 s using a Stomacher 400 Circulator (Seward, West Sussex, UK). Mixed final solutions were then stored at -20°C until DNA extraction.

The DNA from whole rumen content and milk, as well as an internal control sample, was extracted using the double bead-beating plus column (DBB+C) method as described by Zaheer et al. (2018) with modifications. Briefly, aliquots of 350 μL of sample were transferred to a tube containing sterile zirconia beads, 1 mL resuspen-

sion buffer with 0.5% β -mercaptoethanol, and 200 μL of 10% SDS. Cells were disrupted by bead beating using an Omni Bead Ruptor 24 (Perkin Elmer, Shelton, CT) at 4,000 rpm for 3 min, followed by an incubation of 15 min at 70°C (Eppendorf Thermomixer C, Eppendorf, Mississauga, Canada). After centrifugation (15 min at $16,000 \times g$ and 4°C), supernatant was collected and the pellet was submitted to a second round of bead beating, 70°C incubation, and centrifugation. Nucleic acid precipitation was achieved by adding 200 μL of 10 M sodium acetate solution and 1 mL of isopropanol. Pellets were rehydrated in 200 μL of 1 $\times$ tris(hydroxymethyl) aminomethane-ethylenediaminetetraacetic acid before overnight incubation at 4°C. For milk samples, 5 μL of GlycoBlue (Burlington, CA) was added as a co-precipitant before the addition of isopropanol.

The RNA was then removed by adding RNase A (ThermoFisher Scientific, Waltham, MA) followed by incubation at 37°C for 15 min. After Proteinase K treatment, DNA was purified using the QIAamp Fast DNA Stool Mini Kit (Qiagen, Germantown, MD) according to the manufacturer's instructions. The DNA from both rounds of bead beating from the same sample was pooled when loaded on the purification column. The DNA concentrations and optical density, which are measured as the spectrophotometric absorbance at nucleic acid peak to protein absorption peak (260/280) for protein contamination and nucleic acid peak (260/230) to contamination peak to assess purity. These were determined using a NanoDrop 8000 spectrophotometer (ThermoFisher Scientific, Waltham, MA). The DNA integrity was verified by agarose (1%) gel electrophoresis.

Sequencing and Bioinformatics

The DNA samples were sent to Genome Quebec for shotgun metagenomic sequencing using the Illumina NovaSeq 6000 sequencing system in paired-end format of 150 bp according to manufacturer's instructions (Illumina, San Diego, CA). Analyses were processed using the AAFC metagenomics_nf Nextflow pipeline, which is available at https://github.com/AAFC-Bioinfo-AAC/metagenomics_nf. All fastq sequences were preprocessed using fastp (v 0.23.4) to remove adapters and low-quality bases. To remove contamination from the host, quality-filtered reads were mapped against a reference genome made of the PhiX and ARS-UCD 1.3 bovine genome using Bowtie2 (v 2.5.1; Langmead and Salzberg, 2012). Reads that did not align to the reference genome were considered microbial and recovered with BEDtools (v. 2.30.0; Quinlan and Hall, 2010). Taxonomy was assigned using 2 different approaches, rumen samples were processed using the Kraken2 (v 2.1.2; Wood et al., 2019) and Bracken (v. 2.9; Lu et al., 2017) branch of the metagenomics_nf workflow and their reads were assigned using Kraken2 and Bracken databases prepared with data from the Genome Taxonomy Database (GTDB r214; Parks et al., 2018). Note that the Kraken2 assignment was done with a confidence score of 0.5 and that the Bracken database was built with the same threshold. In contrast, reads from milk samples were processed with MetaPhlAn (v 4.0; Blanco-Miguez et al., 2022) and assigned to the innate database. Microbial composition of the samples was analyzed and graphed in R (4.4.2; R Foundation for Statistical Computing) using dependencies from the tidyverse (v. 2.0.0) and phyloseq (v. 1.50.0; McMurdie and Holmes, 2013) packages. All sequences have been deposited under BioProject PRJNA1311556 in the National Center for Biotechnology Information Short Read Archive (<https://dataview.ncbi.nlm.nih.gov/object/PRJNA1311556?reviewer=f5tulc0eca5bbrvfib9rf3s7hv>).

De novo assemblies were conducted on individual samples using MEGAHIT with default settings and the --min-contig-len 1000 parameter. Additionally, the metagenomics_nf workflow was run with the co-assembly option enabled, generating 5 co-assemblies, each consisting of a small set of samples (9 to 11) collected at the same time point (d 2, 7, 21, and 28). For these co-assemblies, MEGAHIT was used with the parameters --min-contig-len 1000, --min-count 2, and --k-list 21, 41, 61, 81, 99. In both individual and co-assembly branches of the workflow, prepared reads from each sample were mapped to de novo assemblies using Bowtie2, and sequence coverage was assessed with jgi_summarize_bam_contig_depths from BBTools (Department of Energy–Joint Genome Institute). Contigs were then

binned into metagenome-assembled genomes (MAG) using MetaBAT2 (v 2.2.15; Kang et al., 2019), and their quality was evaluated with CheckM2 (v 1.0.1; Chklovski et al., 2023). The MAG with >90% completeness and <5% contamination were retained. Regardless of whether they originated from individual or co-assemblies, MAG were de-replicated on a per-sample basis using dRep (v 2.4.3; Olm et al., 2017) to remove redundancy, ensuring that only one representative MAG was retained per genome set. Only 5 MAG were able to be recovered from milk samples, and as such, these have been omitted from the rest of the analysis due to the lack of depth, and functional profiling was completed on rumen samples only.

The taxonomy of MAG from rumen samples was assigned using GTDB-Tk (v 2.1.1) with the GTDB release r214.2. Functional profiling of the samples was predicted using DRAM (v 1.4.6; Shaffer et al., 2020) with a total of 402 MAG as input. Briefly, metabolic pathway data including completeness scores and presence or absence information for 99 pathways was divided into 2 separate data frames: one containing numerical completeness scores and another with binary presence or absence values. Boolean presence or absence values were converted to numeric format (0 or 1) to facilitate statistical analyses. Mean pathway completeness scores were then computed for each group to enable comparative analyses. To identify metabolic pathways differentially represented across treatment groups, linear discriminant analysis effect size (LEfSe) was conducted using the original Python implementation (Segata et al., 2011). Heatmaps were created using an in-house script to compare metabolic pathways to microbial groups (Supplemental File S1, see Notes).

Statistics

Statistical analyses were conducted on DMI, rumen fermentation parameters, milk production and components, and microbial families from rumen samples using a mixed model design with repeated measures in SAS (release 9.4, SAS Institute Inc., Cary, NC). Data were analyzed for fixed effects of sampling day, dietary treatment (MP80 vs. MP100) and their interaction, through the repeated effects of day with cow as the subject and an autoregressive heterogeneous covariance structure to model the correlation among repeated measures over time. Slicing was applied to the diet by day interactions and orthogonal contrast statements were used to test significant difference between day and diets within phases, and their interaction creating 4 groups (PP100 vs. PP80 and RP100 vs. RP80). Significance was declared with $P \leq 0.05$ and trends with $0.05 \leq P \leq 0.10$. Spearman correlations were done for assessing the relationships between rumen microbiota and the differentially abundant microbial pathways in the rumen, as well as between rumen

Table 1. Performance, milk composition, and milk fatty acids compared between treatments and days¹

Parameter	MP80				MP100				SEM	<i>P</i> -value		
	d 2	d 7	d 21	d 28	d 2	d 7	d 21	d 28		Diet	Day	Diet × Day ²
Performance												
DMI, kg/d	16.51	17.64	24.27	24.56	16.08	20.52	21.35	22.82	1.465	0.74	<0.0001	0.01
NH ₃ , mmol/L	12.76	7.30	19.67	21.21	17.96	12.12	18.65	17.22	2.298	0.50	<0.0001	0.13
Rumen pH	6.15	5.91	5.64	5.73	6.05	5.60	5.69	5.62	0.103	0.24	0.004	0.06
Milk production, kg/d	20.48	33.33	42.54	44.10	20.60	36.36	43.00	44.22	2.640	0.77	<0.0001	0.51
SCC, ×1,000	388.80	95.80	43.60	49.00	324.33	62.53	22.00	24.33	82.182	0.58	0.04	0.99
Milk composition												
TS, %	14.55	12.99	12.64	12.64	16.09	13.49	13.11	12.96	0.593	0.22	0.02	0.87
Milk fat, %	2.97	3.98	3.95	4.02	3.83	4.40	4.13	4.18	0.447	0.38	0.22	0.87
Milk protein, %	6.79	3.76	3.24	3.17	7.20	3.84	3.34	3.19	0.560	0.73	<0.0001	0.88
Lactose, %	3.85	4.31	4.50	4.50	3.74	4.36	4.69	4.66	0.073	0.38	<0.0001	0.23
MUN, mg/dL	13.48	1.56	5.74	7.42	16.87	7.08	6.78	6.30	1.866	0.17	<0.0001	<0.0001
BHB	0.02	0.13	0.08	0.09	0.11	0.11	0.10	0.12	0.027	0.17	0.01	0.06
Milk FA, ³ mg/g DM												
C14:0	0.75	0.36	0.42	0.45	0.91	0.40	0.35	0.35	0.069	0.89	<0.0001	0.09
C16:0	2.30	1.18	1.21	1.29	2.53	1.25	1.06	1.08	0.162	0.89	<0.0001	0.26
C18:0	0.33	0.57	0.48	0.42	0.38	0.57	0.54	0.48	0.070	0.62	<0.0001	0.75
C18:1	0.63	0.98	0.83	0.79	0.80	1.11	1.07	1.00	0.182	0.38	0.03	0.76
De novo FA	1.14	0.88	1.03	1.09	1.44	1.01	0.90	0.90	0.110	0.77	0.07	0.06
SCFA	0.24	0.37	0.40	0.41	0.29	0.43	0.38	0.37	0.033	0.67	0.01	0.11
MCFA	3.48	1.70	1.86	1.94	4.07	1.84	1.65	1.61	0.285	0.83	<0.0001	0.17
LCFA	0.76	1.51	1.31	1.24	1.03	1.72	1.63	1.57	0.245	0.34	0.002	0.88
Mixed FA	1.74	1.17	1.21	1.31	1.84	1.28	1.08	1.15	0.115	0.81	0.001	0.26

¹MP80: 80% MP fed during the postpartum phase (PP; sampling d 2 and 7) and 100% MP during the recovery phase (RP; sampling d 21 and 28); MP100: fed 100% of MP requirements.

²Significant interactions were tested for splice analysis with the following results. DMI: PP versus RP, $P < 0.001$; RP80 versus RP100, $P = 0.09$; MUN: PP versus RP, $P = 0.03$.

³SCFA = short-chain fatty acid; MCFA = medium-chain fatty acid; LCFA = long-chain fatty acid; FA = fatty acid.

microbiota and milk components. Differential abundance was calculated using the microbiome multivariable associations with linear models (MaAsLin2) package (v 1.15.1; Mallick et al., 2021), with treatments within phase as fixed effects and individual cows as random effects, and with control group during recovery used as the baseline reference. Correlation analysis was conducted using PROC CORR in SAS using Spearman's rank correlation spliced by dietary treatment within phase. Fair correlations were ranked at $0.30 < r < 0.50$, moderate at $0.5 \leq r \leq 0.7$, and strong at $0.70 < r < 0.9$, and substantial at $r > 0.90$.

RESULTS

Production Performance and Rumen Metabolites

Dry matter intake increased after d 2 postpartum, and an effect of day by diet interaction was seen ($P = 0.01$; Table 1); however, a splice analysis did not show an effect of any specific day. As a result, contrast analysis was performed to look at phase and treatment within phase interactions. There was a significant difference between PP and RP for DMI ($P < 0.001$), and a tendency toward an increased DMI for RP80 versus RP100 ($P =$

0.09). Ammonia-N showed no interaction but was lowest on d 7 according to splice analysis ($P = 0.031$). Rumen pH showed a tendency toward interaction between diet and day, with significant contrast between PP and RP samples ($P = 0.002$) and the lowest values on d 7 ($P = 0.024$). Milk production increased, and SCC decreased with days postparturition but there was no effect of diet or any interaction.

Milk TS decreased with increasing milk production over the course of the experiment, but milk fat percentage did not change (Table 1). Although milk protein did not show an effect of diet, it did decrease in percentage from d 2 to 28 ($P < 0.001$); whereas no significant splice was found, a contrast between PP and RP samples was determined ($P < 0.001$). Concentration of lactose increased from d 2 to 21 and then remained constant to d 28 (Table 1). Milk urea nitrogen showed a significant interaction between diet and day, and contrast analysis showed an effect of phase with an average of 9.75 mg/dL in the PP phase compared with 6.56 in the RP ($P < 0.001$), mostly due to high values immediately after parturition. When splice analysis was performed, samples in the RP tended toward variation on d 21 ($P = 0.089$) and were significantly different on d 28 ($P = 0.032$). Interestingly, milk BHB showed a tendency toward an effect of

interaction between day and diet; however, neither slice analysis nor orthogonal contrasts could assign the variation to a specific factor.

Milk FA analysis showed that C14:0 and de novo FA both had a tendency toward an effect of diet by day interaction (Table 1). Specifically, C14:0 and de novo FA were highest immediately after calving, with a significant contrast for C14:0 for PP versus RP ($P < 0.001$) and RP80 versus RP100 ($P = 0.015$). When looking at the splice analysis, C14:0 showed an effect for d 21 ($P = 0.049$). Additionally, C16:0, medium-chain fatty acid (MCFA) and mixed FA all showed an effect of day with values highest immediately after calving all with a contrast effect of PP versus RP ($P < 0.001$). Additionally, C16:0 and MCFA showed a significant contrast between RP80 and RP100, with higher values in the RP80 group ($P < 0.001$).

Taxonomic Profile of Rumen and Milk

The 30 most abundant microbial families in our samples were visualized by diet and phase to observe the potential changes in composition. In the rumen (Figure 2A), taxonomic differences between groups were not visually different within the top 30 families, following near identical patterns between PP and RP, regardless of MP supply. However, the most abundant families were *Bacteroidaceae* and *Lachnospiraceae*, which together comprised slightly less than 50% of the relative abundance across samples, with no effect of diet seen (Table 2; Supplemental Table S1, see Notes). *Lachnospiraceae* did show a tendency toward a diet by day interaction but neither contrast nor slice analysis could identify a clear pattern in the data. Although many other families showed an effect of day, only *Erysipelotrichaceae* showed a significant interaction between diet and day ($P < 0.001$), and *Paludibacteraceae* and *Anaerovoracaceae* showed a similar tendency. Whereas *Anaerovoracaceae* showed no contrast or splice effects, *Paludibacteraceae* showed a trend toward a treatment effect within phase for both PP and RP ($P = 0.07$ and $P = 0.08$, respectively). *Erysipelotrichaceae* had a contrast between MP80 versus MP100 cows in the PP ($P = 0.05$), and slice analysis showed d 7 with the greatest difference between the treatments (0.01% vs. 0.007% for MP80 vs. MP100 cows, respectively; $P = 0.03$). *Lachnospiraceae* showed a tendency toward an interaction between dietary treatment and day ($P = 0.07$), but no contrast effects were observed.

There were 2 families with no significant interaction effect, but an effect of treatment: *Ruminococcaceae* ($P = 0.05$) and *Paludibacteraceae* ($P = 0.04$), both of which were increased in the MP100 dietary treatment. *Ruminococcaceae* showed a significantly reduced relative abundance on d 7 ($P = 0.01$) and contrast analysis showed an

effect of phase ($P < 0.001$) and treatment within phase for the PP80 versus PP100 with an increased relative abundance in PP100 versus PP80 cows ($P = 0.002$). *Treponemataceae* was decreased in relative abundance on d 2 ($P = 0.024$) and showed a contrast between PP80 and PP100 samples ($P = 0.007$). Several additional families showed a significant contrast between PP and RP samples. More abundant groups including *Bacteroidaceae* (25.9% vs. 32.5%), and *Coprobaclillaceae* (0.09% vs. 0.48%) had a lower relative abundances in PP compared with the RP, whereas *Fibrobacteraceae* ($P < 0.001$) was higher immediately after calving compared with d 21 (0.02% vs. 0.002%, respectively).

Diversity analysis of the rumen microbiota as measured by the Shannon index (Table 2) also showed an effect of day with lowest diversity on d 7 postparturition (5.87 *H*, for both dietary treatments). However, contrast analysis showed no effect of phase, and slice analysis showed a difference of treatment on d 2 ($P = 0.04$).

Analysis of the milk microbiome was performed despite lower coverage in reads due to bovine contamination (Supplemental Table S3) to provide an overview of the dominant community members. In comparison, diversity analysis of the milk microbiota (Table 3) showed an effect of diet by day interaction. Contrast analysis indicated an effect of phase (PP vs. RP; $P = 0.008$) and slice analysis characterized the variation showing a trend toward variation on d 7 ($P = 0.08$). The dominant family across all milk samples was *Alcaligenaceae*, specifically the genus *Achromobacter* (Supplemental Table S2, see Notes), accounting for as much as 72% of the population (RP80 and PP80; Figure 2B). Statistical analysis at the family level of taxonomy showed a trend ($P = 0.07$) toward *Alcaligenaceae* being higher in MP80 cows compared with MP100 cows, with no effect of day (Table 3). Contrast analysis showed a tendency toward higher relative abundance of *Alcaligenaceae* in PP80 animals compared with PP100 ($P = 0.057$). *Moraxellaceae* showed a tendency toward an interaction effect of diet and day ($P = 0.09$); however, both splice and contrast analysis were not significant. Although *Micrococcaceae* showed no interaction effect, contrast analysis indicated a difference between PP80 and PP100 animals ($P = 0.05$), and splice analysis a tendency toward an effect at d 7 ($P = 0.058$). *Nitrobacteraceae* showed an effect of day with an increase in relative abundance throughout the experiment ($P = 0.05$), which was supported by contrast analysis showing an effect of phase ($P = 0.014$) and a tendency toward increased relative abundance in PP80 animals compared with PP100 ($P = 0.075$). Although not among the dominant groups, overall analysis of the taxonomic composition showed more similarity during RP across dietary treatment groups, with the exception of *Methanobacteriaceae*, *Hyphomicrobiaceae*, and *Rho-*

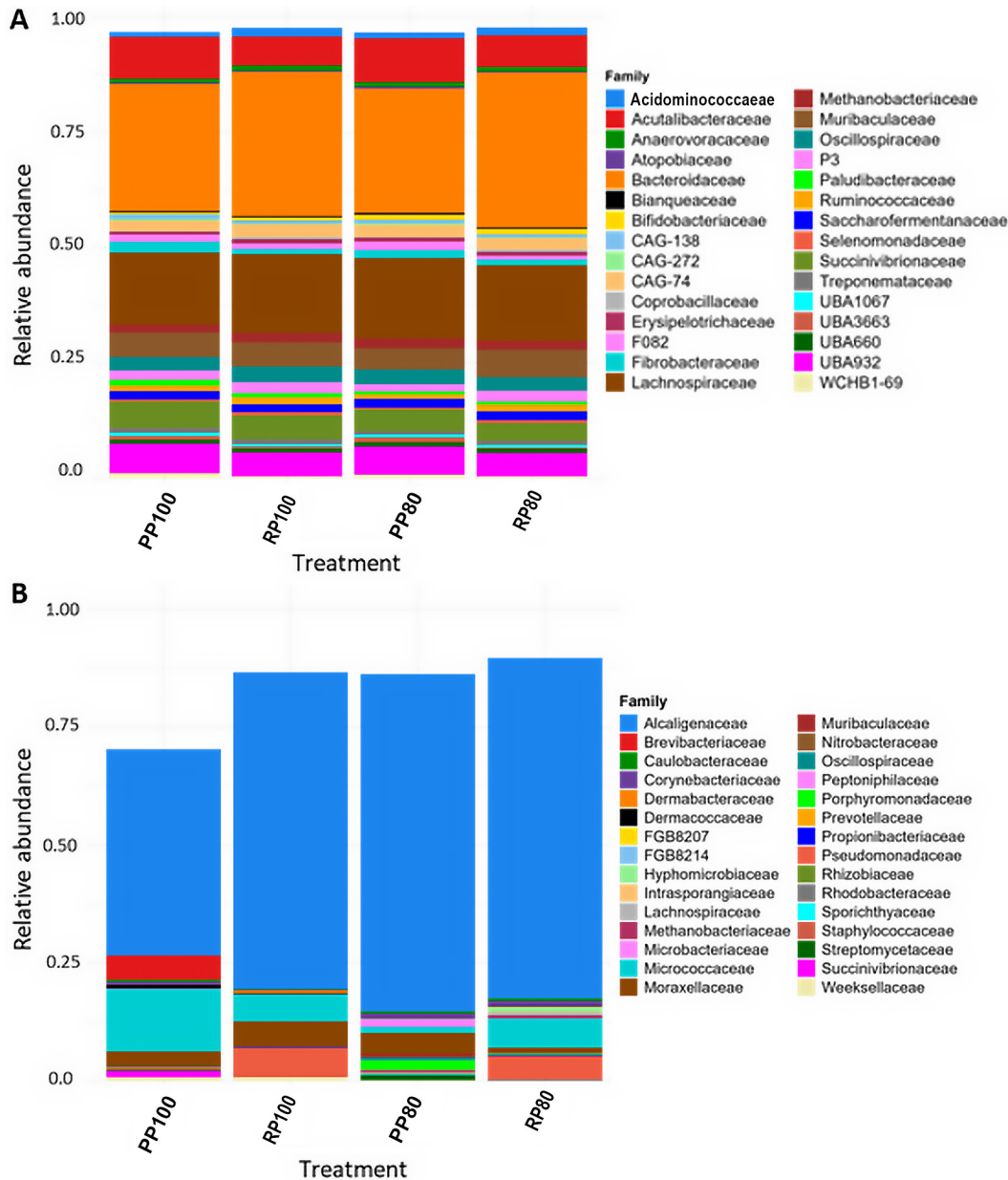


Figure 2. Abundance plot of the top 30 microbial families detected in (A) rumen samples and (B) milk samples. The stacked bars represent the average proportion of each family in the communities of the time-treatment combinations. PP100: animals fed 100% of recommended MP during the first 2 wk postparturition; PP80: animals fed 80% of recommended MP during the PP; RP100: animals fed MP100 during the RP; RP80: animals with dietary change from 80% MP, then converted to MP100 on d 14 and sampled on d 21 and 28 in the RP.

dobacteraceae, which were only detected in the milk of MP80 animals (Figure 2B).

Differential Abundance of Taxa for Rumen and Milk Samples

Differential abundance analysis was performed using the microbiome multivariable associations with linear models analysis to determine dietary treatment within phase differences, using the RP100 cows as a covariate.

In the rumen samples (Figure 3A), the 30 most significantly different families were near identical in both MP80 and MP100 cows during the PP phase. In comparison, during the RP phase, the MP80 cows had no significant differences in taxonomy. The equivalent analysis was completed with milk samples and, similarly, no differences were observed.

To compare the diet effect only during the PP phase, differential abundance analysis was also done using the MP100 cows as the reference value, with just samples

Table 2. Diversity indices and relative abundance of taxonomic family compared between MP80 cows versus MP100 cows and between sampling days in the rumen¹

Parameter	MP80				MP100				SEM	P-value		
	d 2	d 7	d 21	d 28	d 2	d 7	d 21	d 28		Diet	Day	Diet × Day ²
Shannon diversity index (<i>H</i>)	6.37	5.87	6.11	6.08	6.54	5.87	6.24	6.11	0.138	0.54	<0.001	0.88
Taxonomy, %												
<i>Acidominococcaceae</i>	0.012	0.014	0.017	0.016	0.009	0.011	0.021	0.017	0.0017	0.83	<0.001	0.11
<i>Acutalibacteraceae</i>	8.49	8.60	6.35	7.36	9.29	8.90	5.83	6.72	1.854	0.99	0.09	0.93
<i>Anaerovoracaceae</i>	0.011	0.010	0.010	0.010	0.010	0.009	0.014	0.009	0.0011	0.54	0.14	0.76
<i>Atopobiaceae</i>	0.30	0.43	0.28	0.20	0.25	0.24	0.25	0.21	0.113	0.61	0.06	0.10
<i>Bacteroidaceae</i>	25.6	23.0	34.2	33.0	26.5	28.7	30.0	32.6	2.85	0.83	<0.001	0.29
<i>Bianqueaceae</i>	0.39	0.25	0.29	0.27	0.34	0.20	0.38	0.22	0.044	0.68	0.00	0.25
<i>Bifidobacteriaceae</i>	25.6	26.5	23.0	28.7	34.3	30.0	33.0	32.6	2.85	0.14	0.16	0.23
CAG-74	0.03	0.03	0.03	0.02	0.02	0.02	0.03	0.02	0.003	0.78	0.02	0.30
<i>Coprobacillaceae</i>	0.06	0.12	0.60	0.39	0.01	0.16	0.68	0.27	0.273	0.96	0.21	0.95
<i>Erysipelotrichaceae</i>	0.007	0.011	0.009	0.008	0.007	0.007	0.011	0.008	0.0009	0.38	<0.001	0.01
<i>Fibrobacteraceae</i>	0.020	0.015	0.002	0.022	0.030	0.018	0.002	0.020	0.0033	0.27	<0.001	0.24
F082	2.51	0.73	0.98	0.69	2.18	0.86	1.51	0.86	0.408	0.73	0.00	0.84
<i>Lachnospiraceae</i>	17.8	13.6	16.5	16.3	14.6	16.5	18.5	15.6	1.53	0.82	0.17	0.07
<i>Methanobacteriaceae</i>	0.023	0.020	0.020	0.019	0.018	0.016	0.024	0.016	0.0027	0.47	0.09	0.20
<i>Muribaculaceae</i>	2.47	5.77	7.35	4.58	3.09	7.57	6.41	4.00	0.880	0.76	<0.001	0.31
<i>Oscillospiraceae</i>	0.032	0.033	0.029	0.029	0.032	0.026	0.041	0.027	0.0038	0.79	0.13	0.14
<i>Paludibacteraceae</i>	0.83	0.37	0.21	0.96	1.41	1.10	0.25	1.49	0.234	0.04	<0.001	0.10
P3	0.015	0.016	0.024	0.024	0.021	0.018	0.025	0.023	0.0027	0.46	0.02	0.48
<i>Ruminococcaceae</i>	0.013	0.006	0.015	0.016	0.015	0.011	0.016	0.015	0.0014	0.05	<0.001	0.15
<i>Saccharofermentanaceae</i>	2.53	0.90	2.01	1.75	2.20	1.37	1.85	1.47	0.261	0.73	<0.001	0.28
<i>Selenomonadaceae</i>	0.003	0.003	0.006	0.007	0.003	0.005	0.007	0.007	0.0012	0.52	0.01	0.58
<i>Succinivibrionaceae</i>	3.48	5.19	2.43	5.22	2.06	9.41	1.57	8.84	1.966	0.72	0.59	0.30
<i>Treponemataceae</i>	0.008	0.004	0.007	0.011	0.014	0.007	0.012	0.010	0.0025	0.14	0.00	0.26
UBA660	1.02	0.64	0.89	0.90	1.00	0.75	0.93	0.76	0.134	0.99	0.18	0.81
UBA932	6.89	4.10	5.29	4.85	8.33	4.44	5.94	4.49	0.725	0.13	0.34	0.72

¹MP80: 80% MP fed during the postpartum phase (PP; sampling d 2 and 7) and 100% MP during the recovery phase (RP; sampling d 21 and 28); MP100: fed 100% of MP requirements.

²Significant interactions were tested using contrast analysis for diet within phase effects; for differences between diet within day effect, a splice analysis was tested as follows. *Erysipelotrichaceae*: PP80 versus PP100, $P = 0.047$, d 7, $P = 0.03$; *Paludibacteraceae*: PP80 versus PP100, $P = 0.07$, RP80 versus RP100, $P = 0.08$; *Ruminococcaceae* PP versus RP, $P < 0.001$, PP80 versus PP100, $P = 0.002$, d 7, $P = 0.01$; *Treponemataceae*: PP80 versus PP100, $P = 0.007$, d 2, $P = 0.024$; *Coprobacillaceae*: PP versus RP, $P = 0.05$; *Bifidobacteriaceae*: PP versus RP, $P = 0.004$; *Selenomonadaceae*: PP versus RP, $P = 0.02$; P3: PP versus RP, $P = 0.03$; *Fibrobacteraceae*: PP versus RP, $P = 0.001$; *Bacteroidaceae* PP versus RP, $P = 0.004$; Shannon: d 2, $P = 0.004$.

Table 3. Diversity indices and relative abundance of taxonomic family compared between MP80 cows versus MP100 cows and between sampling days in the milk¹

Parameter	MP80				MP100				SEM	P-value		
	d 2	d 7	d 21	d 28	d 2	d 7	d 21	d 28		Diet	Day	Diet × Day ²
Shannon diversity index (<i>H</i>)	3.23	1.95	2.46	1.95	2.78	2.94	2.12	2.15	0.28	0.67	<0.001	0.04
Taxonomy, %												
<i>Alcaligenaceae</i>	58.9	84.1	62.5	81.9	33.4	57.7	61.2	72.6	13.27	0.07	0.11	0.46
<i>Dermacoccaceae</i>	0.00	0.00	0.08	0.06	2.33	0.39	0.12	0.36	0.926	0.26	0.77	0.63
<i>Micrococcaceae</i>	2.7	2.4	22.1	2.5	37.3	13.9	9.6	14.0	9.07	0.13	0.45	0.21
<i>Moraxellaceae</i>	17.7	0.3	4.1	0.1	0.9	11.9	24.5	0.3	7.97	0.46	0.01	0.09
<i>Nitrobacteraceae</i>	0.12	0.26	0.19	0.68	0.00	0.00	0.29	0.47	0.171	0.32	0.05	0.69
<i>Porphyrimonadaceae</i>	3.48	0.34	0.18	0.28	0.00	0.00	0.00	0.00	1.872	0.47	0.82	0.82
<i>Weeksellaceae</i>	0.03	0.49	0.15	0.00	0.00	3.62	3.71	0.03	1.735	0.17	0.40	0.25

¹MP80: 80% MP fed during the postpartum phase (PP; sampling d 2 and 7) and 100% MP during the recovery phase (RP; sampling d 21 and 28); MP100: fed 100% of MP requirements.

²Significant interactions were tested using contrast analysis for diet within phase effects; for differences between diet within day effect, a splice analysis was tested as follows. Shannon diversity: PP versus RP, $P = 0.008$, d 7, $P = 0.079$; *Nitrobacteraceae*: PP versus RP, $P = 0.014$, PP80 versus PP100, $P = 0.075$; *Micrococcaceae* PP80 versus PP100, $P = 0.0495$, d 7, $P = 0.0583$, d 28, $P = 0.0458$; *Alcaligenaceae* PP80 versus PP100, $P = 0.0574$, d 7, $P = 0.0957$.

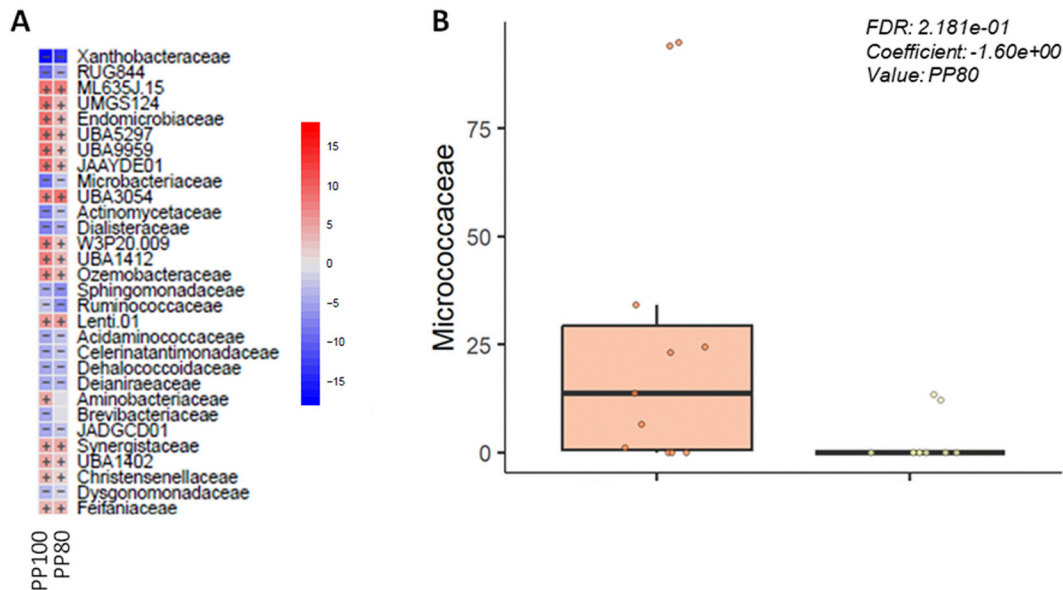


Figure 3. Differential abundance results between time points and treatments. Panel A: heatmap showing the top 30 differentially abundant bacterial families in rumen, with the control group during recovery used as the baseline reference; colors represent the magnitude and direction of the association based on MaAsLin2 model coefficients. Panel B: boxplot displaying the treatment differences during PP for the *Micrococcaceae* family, with individual points jittered across. The center line represents the median of the normalized abundance data. Boxes are the interquartile range, extending from the 25th to the 75th percentile. Whiskers are the range to the smallest and largest values within 1.5 times the interquartile range. PP100: animals fed 100% of recommended MP during the first 2 wk postparturition; PP80: animals fed 80% of recommended MP during the PP; RP100: animals fed MP100 during the RP; RP80: animals with dietary change from 80% MP, then converted to MP100 on d 14 and sampled on d 21 and 28 in the RP. FDR = false discovery rate.

from the MP80 cows. The rumen samples yielded no significant differences; however, in the milk, the family of *Micrococcaceae* was significantly lower in the MP80 compared with the MP100 cows (Figure 3B).

Functional Profiling of the Rumen

As previously stated, insufficient sequencing depth in the milk samples precluded functional profiling via MAG. Therefore, analysis was only conducted using rumen samples. To better understand metabolic differences based on nutrient availability in the rumen, functional pathways were determined from MAG abundances between diets (MP80 vs. MP100) for each sampling day within phases (PP vs. RP). The 4 treatment groups exhibited very similar metabolism profiles, though some differences were observed (Figure 4). Monomethylamine conversion to ammonia was a pathway uniquely partially detected in the MP80 cows during the PP. The pathway for nitrogen conversion to ammonia for MP80 cows had a greater abundance during PP than the RP; however, the pathway “acetate to methane, pt 1,” representing a specific set of steps in this pathway, was less present in comparison. The conversion of nitrate to nitrite was only detected in the RP.

Correlation Between the Rumen Microbiota and Predicted Functional Pathway Abundance

Taxonomies assigned to each MAG were used to look at the predictive microbial involvement in the metabolome. Pathways of interest were determined as those that were observed to have an effect when performing LEfSe using the microbial family as the discriminant feature (Supplemental Figure S1, see Notes). Based on this analysis, key pathways involving SCFA conversion and methanogenesis were then used to perform a correlation analysis, spliced by dietary treatment within phase, to associate them with the presence of specific microbial families (Supplemental Figure S2, see Notes). Specifically, in the PP, there were fewer clear associations between microbiota and associated pathways. In the MP100 cows, UBA660, a partially characterized *Bacillales* member of the *Firmicutes* phylum, was strongly negatively associated with acetate production, and the movement of acetate to methane ($r > -0.72$; $P \leq 0.005$). Whereas, in the MP80 cows, no strong correlations ($-0.70 > r > 0.70$) were found. Most associations were seen in RP, with correlations primarily associated with SCFA production (acetate, butyrate, L-lactate, and formate pathways), as well as the production of ammonia from nitrogen. In the

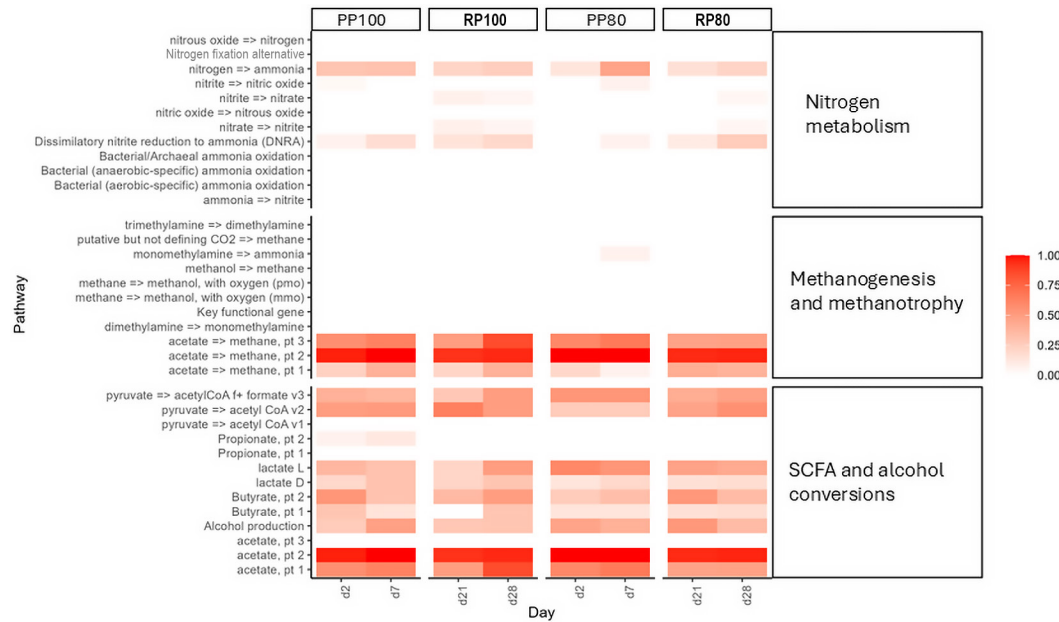


Figure 4. Heatmap displaying the functional pathway abundances in the rumen across treatments and time points, highlighting nitrogen metabolism, methanogenesis, and SCFA and alcohol conversions. For the PP (d 2 and 7) and the RP (d 21 and 28), samples are shown by day within treatment. PP100: animals fed 100% of recommended MP during the first 2 wk postparturition; PP80: animals fed 80% of recommended MP during the PP; RP100: animals fed MP100 during the RP; RP80: animals with dietary change from 80% MP, then converted to MP100 on d 14 and sampled on d 21 and 28 in the RP.

MP80 cows, more positive associations were seen than in the other diet by phase groups, and all positive associations were linked with the family *Ruminococcaceae* ($0.76 < r < 0.81$; $P \leq 0.01$). Interestingly, these were for different pathways compared with MP100 cows during RP including negative correlations for the reduction of nitrite to ammonia, D-lactate production, and pyruvate conversion to acetyl CoA. Overall, 5 families were found to be substantially negatively correlated with pyruvate conversion including UBA660, *Erysipelotrichaceae*, *Acidaminococcaceae*, *Anaerovoracaceae*, and *Bianqueaceae* ($r < -0.90$; $P \leq 0.003$), with *Bianqueaceae* and *Anaerovoracaceae* both having substantial negative correlations to L-lactate pathways as well ($P \leq 0.001$).

Correlations Between Milk Components and the Rumen Microbiota

Spearman rank correlation analysis was conducted to assess possible linkages between the rumen microbiota and milk production and composition, regardless of treatment (Figure 5). Unlike the predicted pathway analysis, substantial correlations were not found; however, some moderate ($0.45 < r < 0.69$) and strong correlations were observed. Rumen family *Acidaminococcaceae* was positively correlated with total milk production ($r > 0.52$; $P \leq 0.001$) as well as the lactose content of the milk ($r > 0.74$; $P \leq 0.001$), whereas rumen *Fibrobacteraceae* was

negatively correlated with these same parameters ($r > -0.35$; $P \leq 0.05$ and $r > -0.48$; $P \leq 0.01$, respectively). In contrast, rumen *Fibrobacteraceae* abundance was positively correlated with the mixed FA, C16:0–2, and de novo FA in the milk, and positively correlated with the SCC ($r > 0.42$; $P \leq 0.01$). In comparison, the relative abundance of rumen-associated *Ruminococcaceae* was negatively correlated with SCC ($r > -0.35$; $P \leq 0.05$). *Selenomonadaceae* was positively associated with lactose content in the milk ($r > 0.60$; $P \leq 0.001$). Inversely, *Acidaminococcaceae* and DMI were moderately negatively correlated with the protein content of milk across all treatments ($r > -0.61$; $P \leq 0.001$ and $r > -0.63$; $P \leq 0.001$, respectively).

DISCUSSION

We evaluated the impact of a 20% reduction in MP supply to dairy cows during the transition period (d –14 to 14 relative to calving), focusing on milk production, composition, and the microbial communities of the rumen and the milk. Using metagenomics, we assessed both microbial ecosystems to better understand how dietary changes can affect the whole animal system. Previous studies have linked the bovine gut microbiota to production traits throughout lactation (Monteiro et al., 2022) and to immune modulation (Zhang et al., 2025). Therefore, our objective was to deepen our un-

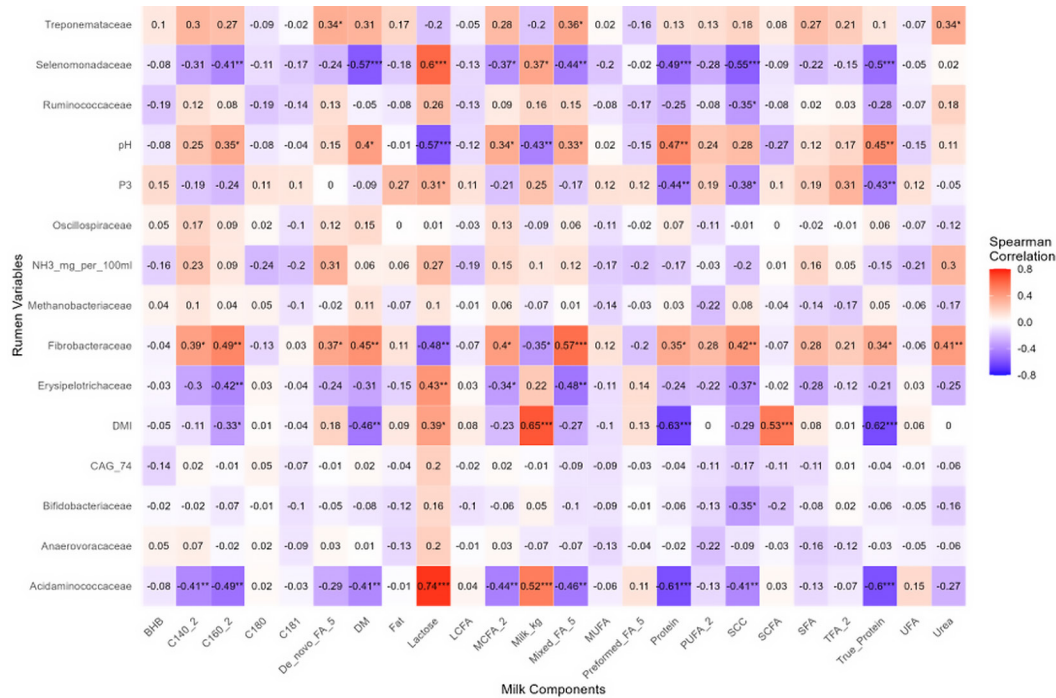


Figure 5. Heatmap displaying Spearman correlation coefficients between DMI, rumen pH, NH_3 , and rumen bacterial families (rows) and milk components, fatty acid profiles, and health indicators (columns). Positive correlations are shown in red, negative correlations in purple, with asterisks denoting statistical significance (* $P \leq 0.05$, ** $P \leq 0.01$, and *** $P \leq 0.001$). SCFA = short-chain fatty acids; MCFA = medium-chain fatty acid; LCFA = long-chain fatty acid; FA = fatty acid; TFA = *trans* fatty acid.

derstanding of the role of dietary change on digestion, metabolism, and rumen and milk microbial community composition as indicators of animal performance during the transition period.

We hypothesized that reducing dietary MP during early lactation would improve rumen microbial efficiency and energy utilization, resulting in improved nitrogen utilization. As previously published (Tapp et al., 2025), the MP-restricted diet did not affect cow BW. However, milk production of the MP80 cows tended to be less than that of the MP100 cows during the period of MP restriction and a diet by day interaction was seen for DMI. Specifically, a tendency toward higher DMI was seen in the MP80 cows compared with the MP100 cows. In the present study, however, no significant changes were observed in the 30 most prominent rumen microbial families between the MP80 and MP100 diets, suggesting that rumen composition remained relatively stable. Given the similar DMI between treatment and the isoenergetic diets, minimal differences in microbiota were likely driven by variations in eating patterns and calving-related stress. However, it is necessary to consider that due to a lack of sufficient rumen genomes in the databases available, the identification of rumen bacteria to genus and species levels is a limitation in being able to adequately interpret the findings of a study such as this.

In the milk microbiota, visual differences between treatments were noted, but multivariate analysis identified only the *Micrococcaceae* family, specifically the genus *Kocuria*, as significantly more abundant in the MP100 cows and significantly increased in the PP. When assessing the variation between days, splice analysis showed variation associated with d 7, whereas numerically d 2 was also higher in relative abundance in the MP100 animals. Although these variations cannot clearly be linked to dietary treatment due to high variation within the dataset, *Kocuria* is considered an opportunistic pathogen, typically found in the environment, on mammalian skin (Savini et al., 2010; Shinozuka et al., 2021), and in bovine milk (Patangia et al., 2023; Shinozuka et al., 2021) and is therefore of great interest. *Kocuria* species have been previously implicated in human infections, can exhibit antibiotic resistance (Kandi et al., 2016), and are therefore of extreme interest within the context of One-Health. Further studies should look at the metabolic relationship of milk-associated strains of *Kocuria* to dietary protein content. The only taxonomic family that showed a potential impact of dietary protein on the relative abundance was *Alcaligenaceae*, which was predominated by the genus *Achromobacter*. While often found in raw milk, these bacteria are considered spoilage organisms that produce proteases and lipases (Niamsiri and Batt, 2009).

Although this does not clearly explain the treatment effect and the specific strain of *Achromobacter* is not known, it is important for future studies to look at these microbes because they are an issue for food safety. Despite the presence of these microbes, and other microbes found within the samples, it remains unclear whether they are transient members of the milk environment or have more persistent roles in association with the udder.

We proposed that restoring MP to 100% of recommendations 2 wk postpartum, alongside increased DMI, would facilitate the transition of rumen microbiota and milk microbiota to their full metabolic potential. No residual effects on milk production were observed during the RP (d 21 and 28), which was anticipated given previous research findings (Tapp et al., 2025). Differential abundance analyses also revealed no lasting differences in microbial abundances between the previously restricted MP80 and the MP100 cows, indicating a full system return to normal. These findings align with those of Zang et al. (2021), in which they found no differences in feed efficiency, milk yield or composition, or health metrics across varying MP ratios in transition cows. In contrast, however, we did observe a tendency for the MP80 cows to consume a greater DMI than MP100 cows during the RP. Our results suggest that a temporary MP restriction may reduce feed costs and nitrogen waste without compromising long-term productivity. Strategic protein management could reduce metabolic stress, nitrogen excretion, and the energy burden of ammonia detoxification for the animal, potentially lowering the risk of having a NEB, ketosis, or fatty liver during this critical transition time. Our predictive pathway analysis showed an increase in monomethylamine to ammonia conversion in the rumen, indicating that there are changes in metabolic balance at MP80. In addition, energy spared from unnecessary protein metabolism could then be redirected toward the immune system function as well, potentially improving resistance to infections such as mastitis (Bronzo et al., 2020). Further research should include the impact of MP reduction on inflammation parameters and nitrogen excretion to validate these assumptions.

Additionally, we explored associations between rumen microbial families, metabolic pathways, and milk components to assess the transition animal using an omics approach to whole-systems biology. The *Acidaminococcaceae* family, known for being able to digest complex polysaccharides and using amino acids (Ryazanov et al., 2021), was positively correlated with total milk production. *Fibrobacteraceae*, a key cellulose degrader in the rumen (Pinnell et al., 2022), was positively correlated with mixed fatty acids, C16:0-2, and de novo FA in the milk, and a moderate correlation to the SCC. This family also showed opposite correlations than *Ruminococcaceae*, which was negatively correlated with SCC, and is

linked to SCFA production and low SCC levels (Bobbo et al., 2020). Although both *Fibrobacteraceae* and *Ruminococcaceae* are fiber-degrading microbes, their contrasting correlations to SCC may reflect differences in their specific roles within the rumen, competitive exclusion, or interactions with host immunity. *Selenomonadaceae* was strongly correlated with milk lactose in the current study and has also been previously reported to produce propionate from succinate generated by *Fibrobacteraceae* (Chuang et al., 2021), indicating a potential syntrophic relationship that may shape rumen fermentation profiles. Although many of these associations were found to only be moderate, it is critical to assess the impacts of nutrition on the metabolic output of dairy cows to better understand how to support the transition cow through the metabolic and physiological changes associated with calving and lactation. By searching for dietary modifications to support rumen and udder health as proxies for animal health, we are able to support production without the need for medical interventions for diet related illnesses.

CONCLUSIONS

Although further research is needed to assess the nitrogen excretion and impacts on inflammation parameters in early-lactation cows given a 2 wk reduction in MP, the minimal taxonomic shifts and stable microbial functions observed in this study suggest that temporary MP restriction is a viable nutritional strategy to enhance nitrogen efficiency without compromising cow health or productivity.

NOTES

This research was financially supported by Agriculture and Agri-Food Canada (Ottawa, ON, Canada). The authors acknowledge the support of the staff of the Sherbrooke Research and Development Centre dairy unit (Sherbrooke, QC, Canada). Specifically, we thank Danielle Bournival, Catherine Thibault, Frederic Beaudoin, and Gabrielle Tapp for their technical assistance. The authors appreciate the statistical support of Steve Méthot (Agriculture and AgriFood Canada, Sherbrooke, Canada), and as well, the support of the dairy barn staff for taking care of the animals. Supplemental material for this article is available at <https://doi.org/10.17632/47xjj8m4js.1> (Supplemental Figures S1 and S2, Tables S1 and S2), <https://doi.org/10.17632/hb8zdp6bkc.1> (Supplemental Table S3, list of samples sequenced, raw read numbers, and read numbers after decontamination of Bovine DNA), and <https://doi.org/10.17632/ghy98rz5xr.1> (Supplemental File S1). Author contributions were as follows: RMP, DO, and CS planned and conducted the experiment. CS obtained the funding. GD conducted analysis. RMP conducted statisti-

cal analysis and conceptualized the manuscript. RMP, JS, and JSB conducted the bioinformatic analysis and graphic development. CS, GD, JS, JSB, and DO reviewed and approved the manuscript. This experiment was conducted in accordance with the guidelines of the Canadian Council on Animal Care (1993) and was approved by Sherbrooke Research and Development Centre Animal Care Committee (no. 591) prior to experimentation. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: FA = fatty acid; LCFA = long-chain fatty acid; LEfSe = linear discriminant analysis effect size; MAG = metagenome-assembled genomes; MCFA = medium-chain fatty acid; MP80 = diet providing 80% of MP requirement; MP100 = diet providing 100% of MP requirement; PP = postpartum phase (d 1–14); PP80 = animals fed 80% of recommended MP during the PP (d 1–14); PP100 = animals fed 100% of recommended MP during the first 2 wk postparturition; NEB = negative energy balance; RP = recovery phase (d 21–28); RP80 = animals with dietary change from 80% MP, then converted to MP100 on d 14 and sampled on d 21 and 28 in the RP; RP100 = animals fed MP100 during the RP; SCFA = short-chain fatty acid; TFA = *trans* fatty acid.

REFERENCES

- Bauman, D. E., and W. B. Currie. 1980. Partitioning of nutrients during pregnancy and lactation: a review of mechanisms involving homeostasis and homeorhesis. *J. Dairy Sci.* 63:1514–1529. [https://doi.org/10.3168/jds.S0022-0302\(80\)83111-0](https://doi.org/10.3168/jds.S0022-0302(80)83111-0).
- Belanche, A., M. Doreau, J. E. Edwards, J. M. Moorby, E. Pinloche, and C. J. Newbold. 2012. Shifts in the rumen microbiota due to the type of carbohydrate and level of protein ingested by dairy cattle are associated with changes in rumen fermentation. *J. Nutr.* 142:1684–1692. <https://doi.org/10.3945/jn.112.159574>.
- Bergman, E. N. 1990. Energy contributions of volatile fatty acids from the gastrointestinal tract in various species. *Physiol. Rev.* 70:567–590. <https://doi.org/10.1152/physrev.1990.70.2.567>.
- Blanco-Míguez, A., F. Beghini, F. Cumbo, L. J. McIver, K. N. Thompson, M. Zolfo, P. Manghi, L. Dubois, K. D. Huang, A. M. Thomas, G. Piccinno, E. Piperni, M. Punčochář, M. Valles-Colomer, A. Tett, F. Giordano, R. Davies, J. Wolf, S. E. Berry, T. D. Spector, E. A. Franzosa, E. Pasolli, F. Asnicar, C. Huttenhower, and N. Segata. 2022. Extending and improving metagenomic taxonomic profiling with uncharacterized species with MetaPhlAn 4. <https://doi.org/10.1101/2022.08.22.504593>.
- Bobbo, T., M. Penasa, and M. Cassandro. 2020. Genetic parameters of bovine milk fatty acid profile, yield, composition, total and differential somatic cell count. *Animals (Basel)* 10:2406. <https://doi.org/10.3390/ani10122406>.
- Brassard, M.-E., P. Y. Chouinard, R. Berthiaume, G. F. Tremblay, R. Gervais, R. Martineau, and D. Cinq-Mars. 2015. Effects of grain source, grain processing, and protein degradability on rumen kinetics and microbial protein synthesis in Boer kids. *J. Anim. Sci.* 93:5355–5366. <https://doi.org/10.2527/jas.2015-9336>.
- Bronzo, V., V. Lopreato, F. Riva, M. Amadori, G. Curone, M. F. Adidis, P. Cremonesi, P. Moroni, E. Trevisi, and B. Castiglioni. 2020. The role of innate immune response and microbiome in resilience of dairy cattle to disease: The mastitis model. *Animals (Basel)* 10:1397. <https://doi.org/10.3390/ani10081397>.
- Canadian Council on Animal Care. 1984. Guide to the Care and Use of Experimental Animals. Vol. 1. Accessed Apr. 22, 2026. https://www.ccac.ca/Documents/Standards/Guidelines/Experimental_Animals_Vol1.pdf.
- Chklovski, A., D. H. Parks, B. J. Woodcroft, and G. W. Tyson. 2023. CheckM2: A rapid, scalable and accurate tool for assessing microbial genome quality using machine learning. *Nat. Methods* 20:1203–1212. <https://doi.org/10.1038/s41592-023-01940-w>.
- Chuang, S.-T., K.-Y. Li, P.-W. Tu, S.-T. Ho, C.-C. Hsu, J.-C. Hsieh, and M.-J. Chen. 2021. Investigating the reciprocal interrelationships among the ruminal microbiota, metabolome, and mastitis in early lactating Holstein dairy cows. *Animals (Basel)* 11:3108. <https://doi.org/10.3390/ani11113108>.
- Jami, E., A. Israel, A. Kotser, and I. Mizrahi. 2013. Exploring the bovine rumen bacterial community from birth to adulthood. *ISME J.* 7:1069–1079. <https://doi.org/10.1038/ismej.2013.2>.
- Jami, E., B. A. White, and I. Mizrahi. 2014. Potential role of the bovine rumen microbiome in modulating milk composition and feed efficiency. *PLoS One* 9:e85423. <https://doi.org/10.1371/journal.pone.0085423>.
- Kandi, V., P. Palange, R. Vaish, A. B. Bhatti, V. Kale, M. R. Kandi, and M. R. Bhoomagiri. 2016. Emerging bacterial infection: Identification and clinical significance of *Kocuria* species. *Cureus* 8:e731. <https://doi.org/10.7759/cureus.731>.
- Kang, D. D., F. Li, E. Kirtan, A. Thomas, R. Egan, H. An, and Z. Wang. 2019. MetaBAT 2: An adaptive binning algorithm for robust and efficient genome reconstruction from metagenome assemblies. *PeerJ* 7:e7359. <https://doi.org/10.7717/peerj.7359>.
- Langmead, B., and S. L. Salzberg. 2012. Fast gapped-read alignment with Bowtie 2. *Nat. Methods* 9:357–359. <https://doi.org/10.1038/nmeth.1923>.
- Leduc, M., R. Gervais, and P. Y. Chouinard. 2023. Nitrogen efficiency in cows fed red clover- or alfalfa-silage-based diets differing in rumen-degradable protein supply. *Animal Open Space* 2:100050. <https://doi.org/10.1016/j.anopes.2023.100050>.
- Lima, F. S., G. Oikonomou, S. F. Lima, M. L. S. Bicalho, E. K. Ganda, J. C. De Oliveira Filho, G. Lorenzo, P. Trojacanec, and R. C. Bicalho. 2015. Prepartum and postpartum rumen fluid microbiomes: Characterization and correlation with production traits in dairy cows. *Appl. Environ. Microbiol.* 81:1327–1337. <https://doi.org/10.1128/AEM.03138-14>.
- Lu, J., F. P. Breitwieser, P. Thielen, and S. L. Salzberg. 2017. Bracken: Estimating species abundance in metagenomics data. *PeerJ Comput. Sci.* 3:e104. <https://doi.org/10.7717/peerj-cs.104>.
- Mallick, H., A. Rahnavard, L. J. McIver, S. Ma, Y. Zhang, L. H. Nguyen, T. L. Tickle, G. Weingart, B. Ren, E. H. Schwager, S. Chatterjee, K. N. Thompson, J. E. Wilkinson, A. Subramanian, Y. Lu, L. Waldron, J. N. Paulson, E. A. Franzosa, H. C. Bravo, and C. Huttenhower. 2021. Multivariable association discovery in population-scale metagenomics studies. *PLOS Comput. Biol.* 17:e1009442. <https://doi.org/10.1371/journal.pcbi.1009442>.
- McArt, J. A. A., D. V. Nydam, G. R. Oetzel, T. R. Overton, and P. A. Ospina. 2013. Elevated non-esterified fatty acids and β -hydroxybutyrate and their association with transition dairy cow performance. *Vet. J.* 198:560–570. <https://doi.org/10.1016/j.tvjl.2013.08.011>.
- McGarvey, J. A., W. G. Miller, S. Sanchez, and L. Stanker. 2004. Identification of bacterial populations in dairy wastewaters by use of 16S rRNA gene sequences and other genetic markers. *Appl. Environ. Microbiol.* 70:4267–4275. <https://doi.org/10.1128/AEM.70.7.4267-4275.2004>.
- McMurdie, P. J., and S. Holmes. 2013. phyloseq: An R package for reproducible interactive analysis and graphics of microbiome census data. *PLoS One* 8:e61217. <https://doi.org/10.1371/journal.pone.0061217>.
- Monteiro, H. F., Z. Zhou, M. S. Gomes, P. M. G. Peixoto, E. C. R. Bon-saglia, I. F. Canisso, B. C. Weimer, and F. S. Lima. 2022. Rumen and lower gut microbiomes relationship with feed efficiency and production traits throughout the lactation of Holstein dairy cows. *Sci. Rep.* 12:4904. <https://doi.org/10.1038/s41598-022-08761-5>.
- Niamsiri, N., and C. A. Batt. 2009. Dairy products. Pages 34–44 in *Encyclopedia of Microbiology*. 3rd ed. M. Schaechter, ed. Academic Press. <https://doi.org/10.1016/B978-012373944-5.00120-6>.

- Olm, M. R., C. T. Brown, B. Brooks, and J. F. Banfield. 2017. dRep: A tool for fast and accurate genomic comparisons that enables improved genome recovery from metagenomes through de-replication. *ISME J.* 11:2864–2868. <https://doi.org/10.1038/ismej.2017.126>.
- Palmonari, A., A. Federiconi, and A. Formigoni. 2024. Animal Board Invited Review: The effect of diet on rumen microbial composition in dairy cows. *Animal* 18:101319. <https://doi.org/10.1016/j.animal.2024.101319>.
- Parks, D. H., M. Chuvochina, D. W. Waite, C. Rinke, A. Skarshewski, P.-A. Chaumeil, and P. Hugenholtz. 2018. A standardized bacterial taxonomy based on genome phylogeny substantially revises the tree of life. *Nat. Biotechnol.* 36:996–1004. <https://doi.org/10.1038/nbt.4229>.
- Patangia, D. V., G. Grimaud, K. Linehan, R. P. Ross, and C. Stanton. 2023. Microbiota and resistome analysis of colostrum and milk from dairy cows treated with and without dry cow therapies. *Antibiotics (Basel)* 12:1315. <https://doi.org/10.3390/antibiotics12081315>.
- Percie du Sert, N., V. Hurst, A. Ahluwalia, S. Alam, M. T. Avey, M. Baker, W. J. Browne, A. Clark, I. C. Cuthill, U. Dirnagl, M. Emerson, P. Garner, S. T. Holgate, D. W. Howells, N. A. Karp, S. E. Lazic, K. Lidster, C. J. MacCallum, M. Macleod, E. J. Pearl, O. H. Petersen, F. Rawle, P. Reynolds, K. Rooney, E. S. Sena, S. D. Silberberg, T. Steckler, and H. Würbel. 2020. The ARRIVE guidelines 2.0: Updated guidelines for reporting animal research. *PLoS Biol.* 18:e3000410. <https://doi.org/10.1371/journal.pbio.3000410>.
- Pinnell, L. J., A. A. Reyes, C. A. Wolfe, M. D. Weinroth, J. L. Metcalf, R. J. Delmore, K. E. Belk, P. S. Morley, and T. E. Engle. 2022. *Bacteroidetes* and *Firmicutes* drive differing microbial diversity and community composition among micro-environments in the bovine rumen. *Front. Vet. Sci.* 9:897996. <https://doi.org/10.3389/fvets.2022.897996>.
- Pitta, D. W., S. Kumar, B. Vecchiarelli, D. J. Shirley, K. Bittinger, L. D. Baker, J. D. Ferguson, and N. Thomsen. 2014. Temporal dynamics in the ruminal microbiome of dairy cows during the transition period. *J. Anim. Sci.* 92:4014–4022. <https://doi.org/10.2527/jas.2014-7621>.
- Plaizier, J. C., M. Danesh Mesgaran, H. Derakhshani, H. Golder, E. Khafipour, J. L. Kleen, I. Lean, J. Loo, G. Penner, and Q. Zebeli. 2018. Review: Enhancing gastrointestinal health in dairy cows. *Animal* 12:s399–s418. <https://doi.org/10.1017/S1751731118001921>.
- Quinlan, A. R., and I. M. Hall. 2010. BEDTools: A flexible suite of utilities for comparing genomic features. *Bioinformatics* 26:841–842. <https://doi.org/10.1093/bioinformatics/btq033>.
- Rehman, A., M. Arif, M. Saeed, A. Manan, A. Al-Sagheer, M. E. A. El-Hack, A. A. Swelum, and A. N. Alowaimier. 2020. Nutrient digestibility, nitrogen excretion, and milk production of mid-lactation Jersey × Friesian cows fed diets containing different proportions of rumen-undegradable protein. *An. Acad. Bras. Cienc.* 92(suppl 1):e20180787. <https://doi.org/10.1590/0001-3765202020180787>.
- Ryazanov, V. A., G. K. Duskaev, S. G. Rakhmatullin, I. S. Miroshnikov, K. P. Miroshnikova, and K. S. Inchagova. 2021. Application of new technologies to assess the effectiveness of feed materials for ruminants. *IOP Conf. Ser. Earth Environ. Sci.* 624:012049. <https://doi.org/10.1088/1755-1315/624/1/012049>.
- Savini, V., C. Catavittello, G. Masciarelli, D. Astolfi, A. Balbinot, A. Bianco, F. Febbo, C. D'Amario, and D. D'Antonio. 2010. Drug sensitivity and clinical impact of members of the genus *Kocuria*. *J. Med. Microbiol.* 59:1395–1402. <https://doi.org/10.1099/jmm.0.021709-0>.
- Segata, N., J. Izard, L. Waldron, D. Gevers, L. Miropolsky, W. S. Garrett, and C. Huttenhower. 2011. Metagenomic biomarker discovery and explanation. *Genome Biol.* 12:R60. <https://doi.org/10.1186/gb-2011-12-6-r60>.
- Shaffer, M., M. A. Borton, B. B. McGivern, A. A. Zayed, S. L. La Rosa, L. M. Solden, P. Liu, A. B. Narrowe, J. Rodríguez-Ramos, B. Bolduc, M. C. Gazitúa, R. A. Daly, G. J. Smith, D. R. Vik, P. B. Pope, M. B. Sullivan, S. Roux, and K. C. Wrighton. 2020. DRAM for distilling microbial metabolism to automate the curation of microbiome function. *Nucleic Acids Res.* 48:8883–8900. <https://doi.org/10.1093/nar/gkaa621>.
- Shinozuka, Y., K. Kawai, T. Kurumisawa, Y. Shimizu, T. Imanishi, A. Ohno, M. Takahashi, S. Kaneko, and N. Suzuki. 2021. Examination of the microbiota of normal cow milk using MinION™ nanopore sequencing. *J. Vet. Med. Sci.* 83:1620–1627. <https://doi.org/10.1292/jvms.21-0353>.
- Tapp, G., P. Lacasse, F. Beaudoin, D. R. Ouellet, F. Malouin, and C. Ster. 2025. Impact of a reduced metabolizable protein supply during the transition period on the immune and metabolic status of dairy cows. *J. Dairy Sci.* 108:2795–2807. <https://doi.org/10.3168/jds.2024-25362>.
- Van Amburgh, M. E., E. A. Collao-Saenz, R. J. Higgs, D. A. Ross, E. B. Recktenwald, E. Raffrenato, L. E. Chase, T. R. Overton, J. K. Mills, and A. Foskolos. 2015. The Cornell Net Carbohydrate and Protein System: Updates to the model and evaluation of version 6.5. *J. Dairy Sci.* 98:6361–6380. <https://doi.org/10.3168/jds.2015-9378>.
- Van Soest, P. 1994. *Nutritional Ecology of the Ruminant*. 2nd ed. Comstock Publishing Associates.
- Weatherburn, M. 1967. Phenol-hypochlorite reaction for determination of ammonia. *Anal. Chem.* 39:971–974. <https://doi.org/10.1021/ac60252a045>.
- Wood, D. E., J. Lu, and B. Langmead. 2019. Improved metagenomic analysis with Kraken 2. *Genome Biol.* 20:257. <https://doi.org/10.1186/s13059-019-1891-0>.
- Zaheer, R., N. Noyes, R. Ortega Polo, S. R. Cook, E. Marinier, G. Van Domselaar, K. E. Belk, P. S. Morley, and T. A. McAllister. 2018. Impact of sequencing depth on the characterization of the microbiome and resistome. *Sci. Rep.* 8:5890–5601. <https://doi.org/10.1038/s41598-018-24280-8>.
- Zang, Y., P. Ji, S. Y. Morrison, Y. Koba, R. J. Grant, and H. M. Dann. 2021. Reducing metabolizable protein supply: Effects on milk production, blood metabolites, and health in early-lactation dairy cows. *J. Dairy Sci.* 104:12443–12458. <https://doi.org/10.3168/jds.2021-20459>.
- Zhang, H., T. Lu, S. Guo, T. He, M.-K. Shin, C. Luo, J. Tong, and Y. Zhang. 2025. Rumen microbes affect the somatic cell counts of dairy cows by modulating glutathione metabolism. *mSystems* 10:e0109324. <https://doi.org/10.1128/msystems.01093-24>.
- Zhou, X. Q., Y. D. Zhang, M. Zhao, T. Zhang, D. Zhu, D. P. Bu, and J. Q. Wang. 2015. Effect of dietary energy source and level on nutrient digestibility, rumen microbial protein synthesis, and milk performance in lactating dairy cows. *J. Dairy Sci.* 98:7209–7217. <https://doi.org/10.3168/jds.2015-9312>.

ORCID

- J. Scott, <https://orcid.org/0009-0004-2554-1564>
 G. Drouin, <https://orcid.org/0009-0001-0153-4499>
 D. R. Ouellet, <https://orcid.org/0000-0002-9835-0767>
 C. Ster, <https://orcid.org/0000-0002-1633-9696>
 R. M. Petri <https://orcid.org/0000-0002-8209-8261>